

Technology Stack

Programming Language

- Python 3.11
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Backend

- FastAPI
- Uvicorn

Purpose:

- REST API development
 - Evaluation endpoint
 - Backend processing
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Frontend

- Streamlit

Purpose:

- User Interface
 - Question submission
 - Display evaluation scores
 - Display retrieved context
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Embedding Model

Sentence Transformers

Model Used:

- all-MiniLM-L6-v2

Purpose:

- Generate vector embeddings
 - Semantic similarity search
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Vector Database

ChromaDB

Purpose:

- Store document embeddings
 - Perform similarity search
 - Retrieve Top-K relevant chunks
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Knowledge Base

Datasets:

- SQuAD v2.0
- TruthfulQA

Purpose:

- Reference answers
 - Retrieval source
-

Data Processing

Libraries:

- Pandas
- JSON

Purpose:

- Dataset preprocessing
 - Chunk creation
-

Evaluation

Techniques Used:

- String similarity
- Keyword overlap
- Retrieval-based verification

Metrics:

- Relevance
- Accuracy
- Hallucination
- Completeness

- Overall Score

PDF Report

Library:

- ReportLab

Purpose:

- Generate evaluation reports

Version Control

- Git
- GitHub

Purpose:

- Source code management

Development Tools

- Visual Studio Code
- PowerShell
- Git Bash

Project Architecture

Frontend

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FastAPI Backend

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Retriever

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ChromaDB

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Knowledge Base



Evaluation Engine



Result Dashboard